Appendix

A Additional Results

A.1 Ablation Results of the Map Construction Modules for Region Classes

Region Precision-Recall for	Precision at Threshold [m]				Recall at Threshold [m]			
	0.5	1.0	2.0	3.0	0.5	1.0	2.0	3.0
RAM	**0.49**	**0.51**	**0.54**	**0.57**	**0.06**	0.13	0.40	0.55
RAM w/o ensemble	0.46	0.48	0.51	0.53	**0.06**	0.16	**0.42**	**0.57**
RAM w/o depth	0.48	**0.51**	**0.54**	0.56	**0.06**	**0.19**	**0.42**	0.56
RAM++	0.44	0.46	0.49	0.53	**0.06**	0.15	**0.42**	**0.57**

Table 4: Tag map ablations evaluated on precision and recall for region classes. For conciseness, crop ensemble filtering is referred to as "ensemble" and depth filtering as "depth".

A.2 Tag Map Precision and Recall for Each Class in Matterport3D

Object Precision-Recall	Precision at Threshold [m]				Recall at Threshold [m]			
	0.1	0.5	1.0	2.0	0.1	0.5	1.0	2.0
bathtub	0.23	0.28	0.30	0.34	0.07	0.24	0.60	0.88
bed	0.53	0.61	0.65	0.69	0.05	0.41	0.68	0.94
cabinet	0.25	0.34	0.41	0.49	0.11	0.45	0.71	0.89
chair	0.39	0.53	0.62	0.72	0.18	0.44	0.63	0.79
chest_of_drawers	0.10	0.17	0.22	0.28	0.13	0.44	0.69	0.86
clothes	0.09	0.13	0.15	0.18	0.07	0.52	0.77	0.90
counter	0.34	0.41	0.46	0.52	0.11	0.36	0.62	0.81
cushion	0.21	0.40	0.51	0.58	0.09	0.36	0.59	0.82
fireplace	0.36	0.50	0.53	0.59	0.09	0.35	0.66	0.88
gym_equipment	0.49	0.60	0.71	0.78	0.06	0.16	0.33	0.65
picture	0.42	0.58	0.68	0.78	0.20	0.49	0.69	0.83
plant	0.26	0.36	0.43	0.51	0.13	0.43	0.64	0.82
seating	0.24	0.32	0.37	0.45	0.03	0.23	0.35	0.51
shower	0.21	0.29	0.35	0.43	0.10	0.40	0.72	0.90
sink	0.18	0.33	0.43	0.51	0.17	0.55	0.81	0.92
sofa	0.30	0.42	0.50	0.56	0.07	0.31	0.50	0.72
stool	0.06	0.09	0.12	0.17	0.15	0.42	0.58	0.76
table	0.30	0.44	0.51	0.61	0.19	0.53	0.73	0.88
toilet	0.45	0.55	0.59	0.67	0.08	0.44	0.74	0.91
towel	0.29	0.41	0.52	0.59	0.03	0.25	0.51	0.67
tv_monitor	0.16	0.31	0.39	0.49	0.17	0.46	0.73	0.86

Table 5: Precision-recall for common object classes evaluated over all 90 scenes of Matterport3D

Region Precision-Recall	Precision at Threshold [m]				Recall at Threshold [m]			
	0.5	1.0	2.0	3.0	0.5	1.0	2.0	3.0
balcony	0.26	0.31	0.38	0.43	0.00	0.00	0.14	0.45
bar	0.00	0.00	0.00	0.00	0.33	0.33	0.33	0.33
bathroom	0.30	0.33	0.35	0.38	0.21	0.38	0.68	0.86
bedroom	0.47	0.49	0.52	0.54	0.06	0.16	0.56	0.85
classroom	0.60	0.60	0.60	0.60	0.00	0.00	0.50	0.50
closet	0.08	0.09	0.13	0.16	0.16	0.23	0.48	0.63
dining room	0.47	0.50	0.53	0.56	0.03	0.16	0.60	0.81
garage	0.69	0.72	0.72	0.77	0.00	0.00	0.29	0.57
hallway	0.54	0.59	0.68	0.73	0.05	0.14	0.43	0.67
kitchen	0.38	0.40	0.42	0.45	0.09	0.29	0.69	0.88
laundryroom/mudroom	0.68	0.70	0.72	0.72	0.00	0.17	0.63	0.89
library	1.00	1.00	1.00	1.00	0.00	0.00	0.50	0.50
living room	0.40	0.42	0.47	0.51	0.07	0.19	0.43	0.61
meetingroom/conferenceroom	0.32	0.36	0.36	0.44	0.04	0.08	0.12	0.28
office	0.36	0.38	0.44	0.48	0.06	0.14	0.37	0.43
porch/terrace/deck/driveway	0.54	0.55	0.58	0.61	0.05	0.11	0.36	0.52
rec/game	1.00	1.00	1.00	1.00	0.00	0.00	0.06	0.06
spa/sauna	0.34	0.37	0.37	0.40	0.02	0.07	0.16	0.23
stairs	0.26	0.29	0.35	0.38	0.08	0.20	0.53	0.72
tv	0.71	0.71	0.82	0.88	0.00	0.31	0.62	0.62
utilityroom/toolroom	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
workout/gym/exercise	0.89	0.89	0.89	0.92	0.00	0.00	0.38	0.75

Table 6: Precision-recall for region classes evaluated over all 90 scenes of Matterport3D

B Tag Map Parameters

The map construction and localization parameters used for all experiments are presented in Table 7 and 8 respectively.

Parameter	Value
Tagging model	ram_swin_large [50]
Crop ensemble border crop percentages	[5%, 10%]
Depth filter mean threshold	0.6 m
Depth filter median threshold	0.6 m

Table 7: Tag map construction parameters

Parameter	Value
Viewpoint near plane distance	0.2 m
Viewpoint far plane distance	80^{th} percentile of depth values
Voxel size	0.2 m
DBSCAN eps	0.4 m
DBSCAN minimum points	5
Normalized votes clustering thresholds	[0.00, 0.25, 0.50, 0.75]

Table 8: Tag map localization parameters

Note that the clustering thresholds on the voxel votes are set relative to the normalized votes, i.e. the voxel votes normalized by the maximum votes of any voxel.